

VEYRAPAY

Digital Payments & Transaction Analytics

Executive Project Report

Smarter Payments. Better Decisions.

Project Type	End-to-end data analytics project
Domain	FinTech / Digital Payments
Data	Purpose-built synthetic transaction data
Period	January 2024 – December 2025
Currency	INR (₹)
Analytics Stack	Excel → Python/Pandas → MySQL/SQL → Power BI/DAX

Portfolio-ready documentation of the completed VeyraPay analytics workflow, validated KPIs, business findings, and executive Power BI dashboard.

1. Executive Summary

VeyraPay is a fictional Indian digital-payments company created as an analytics portfolio project. The project builds an end-to-end workflow around purpose-built synthetic transaction-level data, covering data generation and auditing, Python-based cleaning and exploratory analysis, relational modeling and SQL analysis in MySQL, and an executive Power BI dashboard with DAX-driven KPIs and interactive filters.

The final analytical dataset contains 200,000 cleaned transactions covering January 2024 through December 2025. The completed analysis reports ₹706.50M in transaction value, a 92.01% success rate, 42,103 active customers, and ₹22.76M in refunds. The final Power BI dashboard was validated against Python and MySQL results.

2. Project Overview

The business objective is to provide a unified view of payment performance, transaction reliability, customer behavior, refunds, regional activity, payment methods, and platform usage. The project demonstrates how the same business metrics can be generated, transformed, validated, and presented consistently across multiple analytics tools.

Core business questions

- How are transaction volume and transaction value changing over time?
- Which payment methods contribute the most transaction activity and value?
- What proportion of transactions are successful, failed, or pending?
- How do new and returning customers contribute to activity?
- How significant are refunds in transaction activity and value?
- How does transaction performance vary across regions?
- How do platform and device patterns contribute to payment activity?

3. Objectives

- Build a realistic synthetic digital-payments transaction dataset for India.
- Audit and clean the data while preserving legitimate business-rule missing values.
- Perform exploratory analysis using Python and Pandas.
- Create a structured MySQL analytical model and validate the loaded data.
- Develop SQL business-analysis and KPI-validation queries.
- Build a polished single-page Power BI executive dashboard using DAX.
- Reconcile key outputs across Python, MySQL, and Power BI.
- Produce portfolio-ready technical documentation and GitHub assets.

4. Dataset & Data Architecture

The dataset is purpose-built synthetic data; it is not sourced from Kaggle, GitHub, or real customer/payment records. The source transaction table contains 19 columns covering transaction identity, time, amount, customer attributes, payment method and status, merchant information, geography, platform/device, refunds, and processing time.

Metric	Value
Cleaned transactions	200,000
Source rows before cleaning	200,160
Active customers	42,103
Merchants	6,000
Date dimension	731 dates
Coverage	2024-01-01 to 2025-12-31
Source columns	19

Star-schema design

The analytical model uses fact_transactions supported by dim_date, dim_customer, dim_merchant, dim_payment_method, and dim_geography. Platform and device attributes remain in the transaction fact table; a separate refund fact was not required.

5. Python Data Pipeline

Python/Pandas was used to generate, clean, audit, and analyze the synthetic transaction data. The raw dataset contained 200,160 rows, including 160 exact duplicate rows and 160 duplicate transaction IDs as intentional quality issues. After cleaning, the analytical dataset contained exactly 200,000 rows with zero duplicates.

- Dataset generation produced the raw transaction-level dataset.
- Initial audit documented row counts, columns, distributions, missingness, and duplicate conditions.
- Cleaning removed exact duplicate records while retaining legitimate missing merchant category and city values.
- Exploratory analysis calculated transaction value, status, customer, refund, payment-method, and platform metrics.
- The cleaned dataset and EDA outputs were used as reference points for downstream SQL and Power BI validation.

6. MySQL & SQL Analysis

MySQL was used to create the relational analytical model, populate dimensions and the transaction fact table, and run validation and business-analysis queries. The final fact load contained 200,000 transactions and 200,000 unique transaction IDs.

Table	Rows
fact_transactions	200,000
dim_date	731
dim_merchant	6,000
dim_geography	85
dim_payment_method	5

7. Power BI Executive Dashboard

The final dashboard is a single-page executive performance overview titled “Digital Payments & Transaction Analytics,” covering January 2024 through December 2025. The design uses a dark navy/slate visual system with cyan and teal accents, centered KPI cards, selective charts, and interactive filters.

Dashboard components

- KPI cards: Total Transactions, Total Transaction Value, Success Rate, Active Customers, and Refund Amount.
- Monthly Transaction Value trend.
- Payment Method Performance.
- Transaction Status distribution.
- Transactions by Platform.
- Customer Type comparison.
- Transaction Value by Region.
- Interactive filters for date, region, payment method, and platform.



Figure 1: Final executive dashboard

8. Validated KPI Results

KPI	Validated Result
Total Transactions	200,000
Total Transaction Value	₹706,498,149.45 (₹706.50M)
Successful Transactions	184,024
Success Rate	92.01%
Failed Transactions	11,900 (5.95%)
Pending Transactions	4,076 (2.04%)
Active Customers	42,103
Refund Transactions	10,062 (5.03%)
Refund Amount	₹22,763,481.23 (₹22.76M)

9. Key Business Findings

- UPI is the dominant payment method, with 96,088 transactions and approximately ₹336.84M in transaction value.
- Transaction reliability is broadly consistent across payment methods, with payment-method success rates clustered around 92%.
- Successful transactions account for 92.01% of all transactions, while failed and pending transactions account for 5.95% and 2.04%.
- Mobile is the dominant platform, representing 160,276 transactions (80.14%), compared with 39,724 on Web (19.86%).
- Refund activity totals 10,062 transactions and ₹22.76M in refund amount.
- Regional and customer-type visuals provide additional segmentation for transaction-value and customer-behavior analysis.

10. Data & KPI Validation

Validation was treated as a core part of the project. The final Power BI KPIs and visuals were checked against completed Python EDA and MySQL KPI outputs. Transaction count, transaction value, success rate, active customers, refund amount, status distribution, platform distribution, and payment-method performance were confirmed.

Validation Area	Outcome
Cleaned transaction count	200,000 — matched
Total transaction value	₹706,498,149.45 — matched
Success rate	92.01% — matched
Active customers	42,103 — matched
Refund amount	₹22,763,481.23 — matched
Payment-method results	Matched
Status distribution	Matched
Platform distribution	Matched
Dashboard filters	Functional and checked

11. Limitations & Data Disclosure

- VeyraPay is a fictional company created for portfolio and learning purposes.
- All transaction data is synthetic and purpose-built; it does not represent real customers, merchants, or payments.
- The dataset demonstrates analytics workflows and business reasoning rather than an actual payment processor's production data.
- The project does not claim real-world predictive performance or production operational accuracy.

12. Technology Stack & Project Structure

Layer	Technology / Output
Data generation & analysis	Python, Pandas, NumPy
Data audit	Excel
Database	MySQL
SQL analysis	SQL business analysis & KPI validation
Visualization	Power BI
Analytics expressions	DAX
Version control	GitHub / GitHub Desktop

Repository structure

- **data/** — raw, cleaned, data dictionary, and EDA outputs
- **python/** — dataset generation, cleaning, EDA, and validation scripts
- **sql/** — database setup, data validation, business analysis, and KPI validation
- **powerbi/** — final Power BI report
- **screenshots/** — final dashboard screenshot
- **report/** — project report
- **README.md** — GitHub project documentation
- **.gitignore** — repository ignore rules

13. Conclusion

VeyraPay demonstrates a complete analytics workflow from synthetic data generation through cleaning, exploratory analysis, relational modeling, SQL validation, and executive visualization. The final dashboard converts validated transaction-level data into a concise management view of payment volume, value, reliability, customer activity, refunds, payment methods, platform usage, and regional performance.

The project is structured as a portfolio-ready example of practical data analytics, emphasizing reproducibility, cross-tool KPI consistency, business interpretation, and clear communication of results.

Synthetic-data disclosure: *VeyraPay, its branding, and the transaction dataset are fictional and created for portfolio/educational purposes.*